Experiment No:13
Experiment to handle missing values and inappropriate data in a given data set

AIM:
To handle missing values and inappropriate data in a given data set

DESCRIPTION:
Understand the procedure to handle missing and inappropriate data

CODE:
```
import numpy as np
import pandas as pd

# Read the dataset
df = pd.read_csv("/content/student_data.csv")

# Display the original dataset
print("Original Dataset:")
print(df)

# Check duplicate rows
print("\nDuplicate Rows:")
print(df.duplicated())

# Display dataset information
print("\nDataset Information:")
df.info()

# Remove duplicate rows
df.drop_duplicates(inplace=True)

print("\nDataset after removing duplicates:")
```

```
print(df)

# Reset the index
df.reset_index(drop=True, inplace=True)

# Check the column names
print("\nColumn Names:")
print(df.columns)

# Check unique values in categorical columns
print("\nUnique School Names:")
print(df["school"].unique())

print("\nUnique Gender Values:")
print(df["sex"].unique())

print("\nUnique Address Values:")
print(df["address"].unique())

# Replace inappropriate age values with NaN
df.loc[(df["age"] < 10) | (df["age"] > 25), "age"] = np.nan

# Replace inappropriate grade values with NaN
for col in ["G1", "G2", "G3"]:
    df.loc[(df[col] < 0) | (df[col] > 20), col] = np.nan

# Replace negative absences with NaN
df.loc[df["absences"] < 0, "absences"] = np.nan

# Standardize categorical text
for col in df.select_dtypes(include="object").columns:
    df[col] = df[col].str.strip().str.lower()
```

```
# Fill missing numerical values using the median
for col in df.select_dtypes(include=np.number).columns:
    df[col] = df[col].fillna(df[col].median())

# Fill missing categorical values using the mode
for col in df.select_dtypes(include="object").columns:
    if df[col].isnull().any() and not df[col].mode().empty:
        df[col] = df[col].fillna(df[col].mode()[0])

# Display the final cleaned dataset
print("\nFinal Cleaned Dataset:")
print(df)

# Check for remaining missing values
print("\nMissing Values in Each Column:")
print(df.isnull().sum())

# Display final dataset information
print("\nFinal Dataset Information:")
df.info()

# Save the cleaned dataset
df.to_csv("/content/Cleaned_Student_Dataset.csv",
index=False)

print("\nCleaned dataset saved successfully!")
```

OUTPUT:

Original Dataset:

school sex age address famsize Pstatus Medu Fedu Mjob Fjob \

0	GP	F	18	U	GT3	A	4	4	at_home	teacher	
1	GP	F	17	U	GT3	T	1	1	at_home	other	
2	GP	F	15	U	LE3	T	1	1	at_home	other	
3	GP	F	15	U	GT3	T	4	2	health	services	
4	GP	F	16	U	GT3	T	3	3	other	other	
..	...	..	...	...	...	...	...	...	...	...	
390	MS	M	20	U	LE3	A	2	2	services	services	
391	MS	M	17	U	LE3	T	3	1	services	services	
392	MS	M	21	R	GT3	T	1	1	other	other	
393	MS	M	18	R	LE3	T	3	2	services	other	
394	MS	M	19	U	LE3	T	1	1	other	at_home	

	...	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
0	...	4	3	4	1	1	3	6	5	6	6
1	...	5	3	3	1	1	3	4	5	5	6
2	...	4	3	2	2	3	3	10	7	8	10
3	...	3	2	2	1	1	5	2	15	14	15
4	...	4	3	2	1	2	5	4	6	10	10
..	...	...	...	...	...	...	...	...	..	..	..
390	...	5	5	4	4	5	4	11	9	9	9
391	...	2	4	5	3	4	2	3	14	16	16
392	...	5	5	3	3	3	3	3	10	8	7
393	...	4	4	1	3	4	5	0	11	12	10
394	...	3	2	3	3	3	5	5	8	9	9

[395 rows x 33 columns]

Duplicate Rows:

0	False
1	False

```
2      False
3      False
4      False
      ...
390    False
391    False
392    False
393    False
394    False
Length: 395, dtype: bool

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      395 non-null    object
 1   sex         395 non-null    object
 2   age         395 non-null    int64
 3   address     395 non-null    object
 4   famsize     395 non-null    object
 5   Pstatus     395 non-null    object
 6   Medu        395 non-null    int64
 7   Fedu        395 non-null    int64
 8   Mjob        395 non-null    object
 9   Fjob        395 non-null    object
 10  reason      395 non-null    object
 11  guardian    395 non-null    object
 12  traveltime  395 non-null    int64
```

```
 13  studytime    395 non-null    int64
 14  failures     395 non-null    int64
 15  schoolsup    395 non-null    object
 16  famsup       395 non-null    object
 17  paid         395 non-null    object
 18  activities   395 non-null    object
 19  nursery      395 non-null    object
 20  higher       395 non-null    object
 21  internet     395 non-null    object
 22  romantic     395 non-null    object
 23  famrel       395 non-null    int64
 24  freetime     395 non-null    int64
 25  goout        395 non-null    int64
 26  Dalc         395 non-null    int64
 27  Walc         395 non-null    int64
 28  health       395 non-null    int64
 29  absences     395 non-null    int64
 30  G1           395 non-null    int64
 31  G2           395 non-null    int64
 32  G3           395 non-null    int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB

Dataset after removing duplicates:
   school sex  age address famsize Pstatus  Medu  Fedu      Mjob
Fjob  \
0      GP   F   18       U     GT3       A     4     4   at_home   teacher
1      GP   F   17       U     GT3       T     1     1   at_home     other
2      GP   F   15       U     LE3       T     1     1   at_home     other
3      GP   F   15       U     GT3       T     4     2    health  services
```

```
4      GP   F   16      U     GT3      T     3     3     other     other
..     ...  ..  ...     ...   ...      ...   ...   ...   ...       ...
390    MS   M   20      U     LE3      A     2     2   services  services
391    MS   M   17      U     LE3      T     3     1   services  services
392    MS   M   21      R     GT3      T     1     1     other     other
393    MS   M   18      R     LE3      T     3     2   services     other
394    MS   M   19      U     LE3      T     1     1     other   at_home

     ... famrel freetime  goout  Dalc  Walc health absences  G1  G2  G3
0    ...     4        3      4     1     1      3        6   5   6   6
1    ...     5        3      3     1     1      3        4   5   5   6
2    ...     4        3      2     2     3      3       10   7   8  10
3    ...     3        2      2     1     1      5        2  15  14  15
4    ...     4        3      2     1     2      5        4   6  10  10
..   ...   ...      ...    ...   ...   ...    ...      ...  ..  ..  ..
390  ...     5        5      4     4     5      4       11   9   9   9
391  ...     2        4      5     3     4      2        3  14  16  16
392  ...     5        5      3     3     3      3        3  10   8   7
393  ...     4        4      1     3     4      5        0  11  12  10
394  ...     3        2      3     3     3      5        5   8   9   9

[395 rows x 33 columns]

Column Names:
Index(['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu',
'Fedu',
       'Mjob', 'Fjob', 'reason', 'guardian', 'traveltime', 'studytime',
       'failures', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery',
       'higher', 'internet', 'romantic', 'famrel', 'freetime', 'goout', 'Dalc',
       'Walc', 'health', 'absences', 'G1', 'G2', 'G3'],
```

dtype='object')

Unique School Names:
['GP' 'MS']

Unique Gender Values:
['F' 'M']

Unique Address Values:
['U' 'R']

Final Cleaned Dataset:

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob
0	gp	f	18.0	u	gt3	a	4	4	at_home	teacher
1	gp	f	17.0	u	gt3	t	1	1	at_home	other
2	gp	f	15.0	u	le3	t	1	1	at_home	other
3	gp	f	15.0	u	gt3	t	4	2	health	services
4	gp	f	16.0	u	gt3	t	3	3	other	other
..	...	..	...	...	...	...	...	...	...	...
390	ms	m	20.0	u	le3	a	2	2	services	services
391	ms	m	17.0	u	le3	t	3	1	services	services
392	ms	m	21.0	r	gt3	t	1	1	other	other
393	ms	m	18.0	r	le3	t	3	2	services	other
394	ms	m	19.0	u	le3	t	1	1	other	at_home

	...	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
0	...	4	3	4	1	1	3	6.0	5.0	6.0	6.0
1	...	5	3	3	1	1	3	4.0	5.0	5.0	6.0

```
2   ...     4      3     2     2     3     3    10.0   7.0   8.0  10.0
3   ...     3      2     2     1     1     5     2.0  15.0  14.0  15.0
4   ...     4      3     2     1     2     5     4.0   6.0  10.0  10.0
..  ...    ...     ...   ...   ...   ...   ...     ...   ...   ...   ...
390 ...     5      5     4     4     5     4    11.0   9.0   9.0   9.0
391 ...     2      4     5     3     4     2     3.0  14.0  16.0  16.0
392 ...     5      5     3     3     3     3     3.0  10.0   8.0   7.0
393 ...     4      4     1     3     4     5     0.0  11.0  12.0  10.0
394 ...     3      2     3     3     3     5     5.0   8.0   9.0   9.0

[395 rows x 33 columns]
```

Missing Values in Each Column:

```
school        0
sex           0
age           0
address       0
famsize       0
Pstatus       0
Medu          0
Fedu          0
Mjob          0
Fjob          0
reason        0
guardian      0
traveltime    0
studytime     0
failures      0
schoolsup     0
famsup        0
```

```
paid          0
activities    0
nursery       0
higher        0
internet      0
romantic      0
famrel        0
freetime      0
goout         0
Dalc          0
Walc          0
health        0
absences      0
G1            0
G2            0
G3            0
dtype: int64

Final Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      395 non-null    object
 1   sex         395 non-null    object
 2   age         395 non-null    float64
 3   address     395 non-null    object
 4   famsize     395 non-null    object
 5   Pstatus     395 non-null    object
```

```
 6   Medu         395 non-null    int64
 7   Fedu         395 non-null    int64
 8   Mjob         395 non-null    object
 9   Fjob         395 non-null    object
 10  reason       395 non-null    object
 11  guardian     395 non-null    object
 12  traveltime   395 non-null    int64
 13  studytime    395 non-null    int64
 14  failures     395 non-null    int64
 15  schoolsup    395 non-null    object
 16  famsup       395 non-null    object
 17  paid         395 non-null    object
 18  activities   395 non-null    object
 19  nursery      395 non-null    object
 20  higher       395 non-null    object
 21  internet     395 non-null    object
 22  romantic     395 non-null    object
 23  famrel       395 non-null    int64
 24  freetime     395 non-null    int64
 25  goout        395 non-null    int64
 26  Dalc         395 non-null    int64
 27  Walc         395 non-null    int64
 28  health       395 non-null    int64
 29  absences     395 non-null    float64
 30  G1           395 non-null    float64
 31  G2           395 non-null    float64
 32  G3           395 non-null    float64
dtypes: float64(5), int64(11), object(17)
memory usage: 102.0+ KB
```

Cleaned dataset saved successfully!

RESULT:

Thus, missing values and inappropriate data in the given dataset were successfully identified and handled. Duplicate rows were removed, categorical values were standardized, inappropriate values were replaced with suitable values, and missing numerical and categorical values were filled using the median and mode, respectively. The cleaned dataset was saved successfully as `Cleaned_Student_Dataset.csv`.